

Master Thesis

Decoding Innovation Dynamics in Generative AI: A Patent-Based Topic Modelling and Diffusion Theory Approach

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Abstract

This thesis explores the thematic and temporal dynamics of innovation within the field of Generative Artificial Intelligence (GenAI) using large-scale patent data. In response to the exponential growth of GenAI patent filings over the past decade, the study applies multiple topic modeling algorithms to identify latent technological themes and track their evolution. A tailored preprocessing pipeline was developed to clean and standardize the complex language of patent texts, enhancing the accuracy and interpretability of the extracted topics. The analysis identifies six core innovation themes, reflecting dominant areas of innovation and notable shifts in technological focus. To interpret the diffusion of these technologies, the study applies the Diffusion of Innovations framework, a well-established business theory, offering insights into adoption stages and applicant behavior. Together, these findings offer valuable insights into the emergence, evolution and diffusion of GenAI innovations, supporting more informed decision-making in research, investment, and policy development.

Keywords: Generative AI, Innovation Trends, Patent Analysis, Topic Modelling, Diffusion of Innovations

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1. Introduction

Technological innovation is essential for organizations to gain a competitive edge in today's increasingly interconnected, knowledge-driven global economies. The artificial intelligence (AI) industry plays a pivotal role in this context, reshaping competition across industries. Patents are critical in this process, offering inventors temporary monopolies in return for investments in research and development, particularly in fast-growing domains like digital technologies and AI (Hall & Harhoff, 2012). Among AI subfields, Generative AI (GenAI) stands out for its transformative applications in content creation, design and synthetic data generation (Sengar, Hasan, Kumar, & et al., 2024). Recent advancements, in generative pre-trained transformer (GPT) models, in particular, have enabled near-human performance across tasks, driving innovation across sectors (Dhingra, Singh, SB, Malviya, & Gill, 2023). Reflecting this trend, GenAI patent applications have grown from roughly 700 in 2014 to over 14,000 by 2023, with more than 25% filed in the past year alone (WIPO, 2024). This rapid growth signals both the field's transformative potential and the urgency of understanding its innovation dynamics. Despite this expansion, comprehensive studies that systematically analyze GenAI patent data remain scarce. Existing research often emphasizes citation networks or isolated textual patterns but lacks a thematic and temporal perspective on how innovations evolve over (Rost, 2011; Liu et al., 2021). As a result, there is limited understanding of the shifting technological focus areas and how innovations diffuse across the ecosystem.

This study addresses that gap by applying a comparative topic modeling approach to GenAI patent documents, with the goal of extracting latent themes and tracking their evolution from 2021 onward. The analysis focuses on the content of patent abstracts and claims to identify key innovation trajectories, while excluding legal, ethical and geographic interpretations.

Research Gap and Contribution. Most existing studies on GenAI focus on network analysis or manual bibliometric approaches, which often fail to capture semantic meaning and evolving trends. These methods typically lack adaptability for fast-changing domains. Moreover, few studies apply large-scale text analysis to recent patent data, particularly with theoretical grounding. This thesis addresses these limitations by combining multiple topic modeling algorithms with the Diffusion of Innovations (DoI) framework. This integrated approach reveals how GenAI technologies emerge, mature, and diffuse across sectors. To guide this exploration, the following research questions are formulated:

RQ1: What are the dominant and emerging innovation themes in Generative AI patents filed, and how do these evolve?

RQ2: Which topic modeling algorithms perform best, based on coherence and interpretability, when applied to Generative AI patent data?

RQ3: How can the thematic insights extracted from these patents inform strategic innovation decisions and reveal unexplored opportunities within the Generative AI ecosystem, particularly when framed by the Diffusion of Innovations (DoI) framework?

This research makes three key contributions. First, it identifies dominant and emerging innovation themes in GenAI patents, offering insight into evolving technological focus areas. This contributes to a clearer understanding of the progression and strategic directions of GenAI innovation. Second, it benchmarks topic modeling algorithms, including LDA, LSI, NMF, and BERTopic, based on both quantitative and qualitative performance, offering methodological guidance for future research in patent analytics. Third, by applying the DoI framework, the study maps thematic adoption stages, aiding in the assessment of technological maturity and strategic potential. Together, these contributions offer actionable insights for R&D managers, policymakers and investors aiming to align innovation strategies with emerging GenAI trends. The use of an

international, subject-specific patent dataset from the European Patent Office further enhances the study's novelty and practical relevance.

The thesis is structured as follows to support the investigation of the stated research questions. Chapter 2 reviews relevant literature and theoretical frameworks. Chapter 3 details the research design, including data collection and modeling procedures. Chapter 4 presents the comparative evaluation of topic models. Chapter 5 explores thematic trends and diffusion dynamics. Chapter 6 concludes with key findings, outlines future research directions, and offers a qualitative comparison with recent studies on GenAI patents.

2. Literature Review

The study of technological innovation has utilized various methodologies to analyze advancements and their impacts across different industries. Standard techniques in this field include natural language processing (NLP) methods such as topic modeling, text analysis, large language models (LLMs), and citation and network analysis. These approaches help uncover patterns, structures, and thematic trends within patent and publication data. This review organizes existing research into three core methodological categories: text analysis, topic modeling, and network analysis. It critically examines how each method has been used to study Generative AI (GenAI) and broader AI innovations. The discussion outlines implementation strategies, key findings, and contributions of each method while highlighting ongoing limitations, including issues with model interpretability, the use of outdated data, and methodological fragmentation. Overall, the reviewed literature enhances our understanding of GenAI innovation dynamics while revealing significant gaps for future research.

Topic Modeling

Topic modeling has emerged as a powerful tool for discovering latent themes in collections of innovation data, such as patent documents or academic publications. By clustering frequently co-occurring terms, topic models reveal the major research fronts or application areas without predefined categories (Kherwa & Bansal, 2019). Singh *et al.* (2024) conducted one of the first patent text-mining studies on GenAI, applying Structural Topic Modeling (STM) to 2,398 patents filed between 2017 and 2023. The study aimed to identify key application areas and future development of the project. A key advantage of STM is its ability to incorporate metadata (in this case, patent publication year) to detect temporal shifts in topic prevalence. Similarly, topic modeling was applied to GenAI patents to identify six thematic clusters, ranging from medical applications to cyber-physical systems, which were subsequently mapped through technology roadmaps. The strength of these studies lies in their data-driven taxonomies and the potential for actionable insights for R&D strategy and policy, empowering the audience with valuable information and inspiring them to contribute to this area of research. The use of STM allows for the detection of temporal dynamics and inter-topic correlations, such as overlaps between image generation and conversational agents in multimodal AI. However, both are constrained by static datasets that may not capture post-2023 developments, such as the rising role of GenAI in coding or legal applications. Moreover, the subjectivity in labeling topics and the inherent limitations of unsupervised models remain challenges. On the scholarly side, Dwivedi *et al.* (2023) used STM to analyze AI-related publications in the journal *Technological Forecasting and Social Change (TFSC)*, identifying eight core themes, including AI in healthcare, sustainable supply chains, and AI-driven decision-making. The study offer a valuable cross-disciplinary review of Generative AI, a subset of AI that focuses on creating new content or data, such as images or text, based on existing patterns or examples. However, several limitations reduce its

effectiveness in understanding the dynamics of innovation and strategic application. The framework proposed in the paper is primarily conceptual, relying on high-level taxonomies without empirical validation. While this helps organize existing knowledge, it limits the framework's real-world applicability, particularly in relation to the societal, regulatory, and technological impacts of GenAI. Future research would benefit from grounding such frameworks in quantitative studies or domain-specific case analyses to enhance their practical relevance. Another significant limitation is the omission of the innovation and intellectual property dimension. Although the paper discusses technical capabilities and ethical challenges, it does not engage with patent activity, technological convergence, or innovation diffusion. These are critical for understanding how GenAI technologies emerge and spread across industries. Ignoring these aspects leaves a significant gap in capturing the market dynamics and competitive structures that define the innovation landscape. Furthermore, the framework presented is static and lacks adaptability to rapid developments in the GenAI field. With continual advances in AI models, shifting regulatory environments, and evolving adoption patterns, a fixed taxonomy risks becoming outdated. Incorporating dynamic or longitudinal approaches such as trend monitoring or innovation lifecycle analysis would improve the utility and longevity of such frameworks. Without this adaptability, the model's relevance diminishes in the face of GenAI's fast-paced evolution. Gupta et al. (2024) extended this approach with a systematic review of 1,319 GenAI-related publications from 1985 to 2023. Their use of topic modeling, combined with bibliometric filtering, revealed clusters related to both applications (e.g., image generation) and theoretical concerns (e.g., privacy, GPT-specific research). While offering a comprehensive roadmap for future research, the study's broad timespan raises issues of thematic consistency over time. For example, the meaning of "generative" in early genetic algorithm studies differs significantly from its usage today. Nevertheless, their inclusive data sources and empirically grounded classification strengthen the study's value, motivating further exploration.

Topic modeling has also been applied in related domains of AI innovation. Zhang *et al.* (2021) applied LDA-based topic modeling to blockchain patents, segmenting technological development into lifecycle stages and identifying innovation trends. While methodologically thorough and provides a structured view of technological evolution, its limitations are notable. The use of cosine similarity to measure relationships between topics, when paired with LDA's bag-of-words assumptions, may fail to capture deeper semantic or contextual linkages. Additionally, the dependence on expert-driven parameter tuning for LDA diminishes the method's scalability and objectivity, limiting its automation potential. Despite known limitations, LDA's transparency and reproducibility offer significant advantages over more advanced and less interpretable models.

This study mitigates some of these constraints by combining LDA results with temporal metadata and domain-specific filtering. It leverages the model's strengths to generate actionable insights into the evolution of Generative AI technologies.

Network Analysis and Citation-Based Analysis

Network analysis has long been central to innovation research, especially through co-authorship, co-citation, and patent citation networks. However, traditional techniques are increasingly criticized for their static and descriptive nature. Classic approaches such as main path analysis or modularity-based clustering often rely on binary, rigid relationships (e.g., a patent cites another = 1, else 0) that fail to capture the semantic nuance, directionality of influence, or contextual depth (Duch & Arenas, 2005). Liu *et al.* (2021) constructed a citation network to trace the flow of AI innovation; however, their approach focused solely on structural relationships, overlooking content-based similarities and the influence of strategic citation behavior. This limitation is particularly relevant in the context of Generative AI, where citation patterns may be intentionally distorted, for instance, through limited patent disclosure or the avoidance

of traditional publication channels by companies aiming to protect competitive advantages. Recent evidence from studies on patent-paper pairs suggests that once a patent is granted, the citation rate of the corresponding scientific publication declines, an effect known as the anti-commons phenomenon. In fast-moving domains like Generative AI, citation patterns underrepresent proper knowledge diffusion, especially when innovations are protected by IP or developed in competitive industry settings (Murray & Stern, 2005). More importantly, traditional network analysis is increasingly regarded as methodologically inadequate for capturing the complexity and rapid evolution of technologies such as Generative AI. It treats nodes (e.g., papers, patents, inventors) as isolated units, whereas real-world innovation systems are often multi-level, dynamic, and non-linear. The method assumes static topology even though GenAI collaborations and influence networks shift rapidly, often within months. Additionally, basic network metrics (centrality, degree, clustering coefficient) are descriptive, not explanatory, they reveal who is connected but not why or how that affects innovation outcomes (Saxena & Iyengar, 2020). Network analysis has become a blunt tool in an era that demands finer, more adaptive instruments. Choe et al. (2013) used centrality and topological measures to identify influential patents. However, their exclusive reliance on patents excluded non-patented innovations like open-source contributions, representing a significant portion of technological advancements. Furthermore, the lack of temporal analysis limits understanding how networks evolve. Obreja *et al.* (2024) offered a conceptually nuanced view of AI innovation by combining keyword and citation networks to map ethical and societal discourses. Their study focuses on 1,225 social science articles published between 2000 and 2023. One of their central aims was to uncover non-economic dimensions of AI innovation, such as ethical, cultural, and social factors, that are often secondary to technical and economic analyses. Their analysis reveals a clear shift in the AI innovation discourse, particularly between 2021 and 2023, from a focus on technical performance to ethical governance and

corporate social responsibility (CSR) themes. While “technology” and “performance” continue to dominate the discourse, there has been a noticeable increase in the prominence of terms such as “transformation,” “ethics,” and “CSR.” This shift indicates a growing scholarly interest in how AI-driven change can be guided and managed responsibly, emphasizing social sustainability and ethical considerations. Obreja *et al.* (2024) identified 163 influential papers that explicitly address these non-economic aspects, helping to legitimize emerging discussions around ethical AI, social acceptance, and the human and organizational capabilities required for meaningful innovation in generative AI. Interestingly, even within this socially oriented literature, the technological component still dominates, underscoring the hybrid nature of AI innovation as both a technical and social process. A major contribution of the study lies in its mixed-methods approach: the authors combine bibliometric network mapping (e.g., co-citation and keyword networks) with qualitative analysis. For instance, when a citation cluster forms around themes like “ethical AI governance,” the authors move beyond quantitative metrics to interpret the underlying content, uncovering calls for transparency, the development of new ethical frameworks, and innovative governance models. This qualitative layer adds contextual richness and analytical depth, strengthening their conclusions and avoiding the limitations of a purely data-driven network analysis. However, a key limitation is their focus exclusively on social science literature. By intentionally filtering out economic and technical papers, the study misses how these dimensions interact with ethical or cultural considerations. While rising and falling trends in topics are identified, the underlying evolution of the network itself, such as how new actors reshape the topology or how knowledge flows diversify, is rarely modeled in detail. As the volume of generative AI research grows, there is clear potential for more fine-grained dynamic network approaches. Future work could incorporate simulation or predictive modeling to explore how citation or collaboration networks might evolve, for instance, whether the GenAI

landscape will consolidate around a few dominant actors or become increasingly decentralized and democratized (Krakowski, 2025).

Hybrid and AI-Enhanced Frameworks (LLMs)

To address the limitations of traditional methods, recent work has begun incorporating large language models (LLMs) into innovation mapping. Wang et al. (2024) propose LLM-based semantic knowledge graphs to reveal conceptual similarity between patents, even when citation links are absent. These graphs reveal latent connections, such as methodological overlaps between domains like "music" and "protein" generation, that may facilitate cross-domain innovation. By clustering conceptually related innovations, even when inventors are unaware of each other's work, they offer a more organic and integrated view of the innovation landscape. Toetzke et al. (2023) propose using large language models (LLMs) to automate the monitoring of climate technology innovation across unstructured data. Their framework showed the feasibility of LLM-based classification and mapping but also acknowledged risks like data sparsity and hallucination, especially in prompt-based or zero-shot learning setups. The study provides a compelling vision of scalable innovation monitoring, but its reliance on general-purpose LLMs calls for further validation through domain-specific fine-tuning and transparent evaluation protocols. Zhou et al. (2024) propose a generative AI framework using large language models for patent analysis. By leveraging zero-shot and few-shot prompting, the authors demonstrate how generative AI can extract structured insights from unstructured patent text, perform semantic clustering, classify patents by novelty and domain, and improve topic modeling. Experimental comparisons reveal that LLM-based approaches consistently outperform traditional methods such as TF-IDF + KMeans and LDA and BERT-based models, particularly in semantic alignment and topic coherence. Despite their strong performance, the study identifies several limitations. First, the high computational cost and latency of models like

GPT-4 constrain their scalability in real-world applications. Second, interpretability remains limited, raising legal and regulatory concerns. Third, hallucinations and factual inaccuracies may arise in zero-shot scenarios, potentially undermining the reliability of extracted insights. Finally, domain-specific adaptation continues to be challenging, as general-purpose LLMs may not fully capture the nuanced language of technical or legal patent documents.

Synthesis and Methodological Implications. The reviewed literature reveals that while prior studies have made significant advancements in applying topic modeling, network analysis, and LLM-based tools to innovation research, they often remain methodologically fragmented or focus on outdated data. Network and citation-based analyses emphasize structure over content, limiting their ability to capture evolving themes or contextual depth. Meanwhile, text analysis methods such as STM and LDA have demonstrated promise in identifying thematic clusters but are rarely combined with innovation diffusion frameworks or evaluated for interpretability across models. These gaps directly inform the methodological choices in this thesis: the use of multiple topic modeling algorithms enables both robustness and benchmarking; the integration of patent metadata supports temporal and sectoral mapping; and the adoption of the Diffusion of Innovations (DoI) framework provides a theoretical perspective to interpret how technological themes evolve across time and applicant categories. This combined approach aims to generate both methodological robustness and practical insight into the innovation dynamics of Generative AI.

2.1 Theoretical Framework

This study adopts Rogers' (2003) Diffusion of Innovations (DoI) theory to provide a conceptual foundation for analyzing the innovation patterns in Generative AI (GenAI) patent data. The DoI framework explains

how innovations are communicated and adopted within a social system over time. It has been widely applied in technology adoption studies and is particularly relevant for emerging technologies where development and adoption can differ significantly across actors and contexts. The core idea behind DoI is that innovations diffuse through identifiable stages: knowledge, persuasion, decision, implementation, and confirmation (Rogers, 2003). These stages are reflected in patent activity through filing patterns, the technological focus of patents indicated by their IPC (International Patent Classification) codes, and the involvement of different types of applicants over time. The theory further distinguishes between different adopter types (innovators, early adopters, early majority, late majority, laggards), which can be linked to specific IPC domains or sectors based on the timing and nature of their technological emphasis.

DoI provides a valuable perspective for interpreting topic modeling results, as topics can represent distinct innovation themes that diffuse at different rates (Rogers, 2003). For instance, some IPC-coded technological areas may emerge earlier and be more concentrated among innovative firms or specific applicants. In contrast, others diffuse later or more broadly across sectors and types of applicants. By mapping topics over time and linking them to IPC trends and applicant activity, this study applies the Diffusion of Innovations (DoI) framework to theorize about the structure and evolution of Generative AI innovation. Given the study's exploratory focus on thematic patterns in GenAI patents, the Diffusion of Innovations (DoI) framework is used for interpretation rather than prediction. Accordingly, no dependent variables or hypothesis testing are involved. Instead, the theory supports qualitatively interpreting topic lifecycles and technological diffusion trends.

3. Dataset Construction and Retrieval

GenAI lacks a single dedicated patent classification, making its identification in patent data a non-trivial task (WIPO, 2023). To address this, the study retrieved patents from the Espacenet database (European Patent Office, n.d.) using a carefully designed Boolean search, targeting documents from 2021 to 2025 to capture recent innovations. The search strategy, informed by WIPO (2023), OECD (2020), and the EPO, balanced recall and precision through iterative refinement using Boolean logic, proximity operators, and IPC/CPC classifications. Only the title, abstract, and claims fields were searched to ensure thematic centrality and minimize noise. Following best practices, keyword queries such as “generative* NEAR3 (AI OR artificial intel* OR adversarial)” were combined with domain-specific classifications (e.g., G06N*, G16H*) to ensure relevance. This hybrid strategy leveraged broad keyword coverage while maintaining technical precision through classification filters. While some studies use AI-based semantic filtering (e.g., BERT), this study employed a transparent, rule-based method due to its reproducibility and alignment with Espacenet’s advanced search capabilities.

The query was built around three conceptual blocks: (1) core GenAI architectures, (2) supporting AI terminology and (3) application contexts. Boolean logic and proximity operators were used alongside IPC/CPC classifications (e.g., G06F*, G06T*, G10L*) to improve domain relevance. The query was restricted to the title, abstract, and claims fields for clarity and reduced noise. Classification symbols were validated and refined using tools like the EPO CPC Categoriser, and coverage was focused post-GPT-3 (2021+) to reflect modern GenAI. The search strategy evolved through iterative testing. Initial queries produced false positives related to generative design in unrelated fields. To address this, proximity constraints (e.g., "generative" +1W "AI") and additional keywords (e.g., "diffusion model," "VAE") were introduced.

TIEN ABEN CLEN = ((GENERATIVE* +2W (AI OR "ARTIFICIAL INTELLIGENCE" OR ADVERSARIAL)
 OR (GENERATIVE +2W PRE#TRAINED) OR (PRE#TRAINED +3W ("LANGUAGE MODEL" OR
 TRANSFORMER?)) OR CHAT#GPT OR (VARIATIONAL +2W AUTOENCODER?) OR (GENERATIVE +2W
 ADVERSARIAL +2W NETWORK*) OR (DIFFUSION +2W PROBABILISTIC +2W MODEL?) OR
 (AUTO#REGRESSIV* +2W MODEL*)) OR (GENERATIVE* +3W (AI OR "ARTIFICIAL INTELLIGENCE"
 OR ADVERSARIAL) OR (GENERATIVE +2W PRE#TRAINED) OR (PRE#TRAINED +3W ("LANGUAGE
 MODEL" OR TRANSFORMER?)) OR CHAT#GPT OR (VARIATIONAL +2W AUTOENCODER?) OR
 (CONVOLUTIONAL +2W GENERATIVE +2W ADVERSARIAL +2W NETWORK*) OR (DIFFUSION +2W
 PROBABILISTIC* +2W MODEL?) OR (DIFFUSIONAL +2W NETWORK?) OR (DENOISING +2W
 DIFFUSION +2W PROBABILISTIC +2W MODEL?) OR (GENERATIVE +2W LATENT +2W
 OPTIMIZATION) OR (NEURAL +2W RADIANCE +2W FIELD) OR (AUTO#REGRESSIV* +3W MODEL*)
 OR GAN OR GANS OR GENAI OR VAE OR VAES OR ((DENOISING OR VIDEO OR MODELS OR STABLE)
 +3W DIFFUSION* +2W MODEL*) OR (VARIATIONAL +3W AUTOENCODER*) OR GPT#3* OR
 GPT#4*)
 OR
 (TRANSFORMER* OR AUTOENCODER* OR LLM OR "LARGE LANGUAGE MODEL*" OR GAN OR
 (GENERAT* +3W ADVERSARIAL +3W NETWORK*) OR (AUTO#REGRESSIV* +3W MODEL*) OR
 (DIFFUSION +3W PROBABILISTIC +3W MODEL*))
 OR
 (((IMAGE* OR TEXT* OR VIDEO* OR SPEECH* OR ("3D" +1W MODEL*) OR "GENE SEQ*" OR
 DESIGN OR PROGRAM* OR COMPUTER* OR SOFTWARE* OR CODE* OR MUSIC* OR SCENE* OR
 MOLECULE* OR "SYNTHETIC DATA*" OR "WORD SEQUENCE*" OR "WORD SEQUENCES*") OR
 (GENERATE* OR GENERATING* OR GENERATION* OR GENERATIVE*)) AND (GENERATIVE* +3W
 MODEL*))
 AND
 ((IPC = (G02B* OR G01N* OR B23Q* OR A61B* OR G06F* OR G06N* OR G06Q* OR G06V* OR
 G10L* OR G16H* OR H04L* OR H04N* OR G06T* OR G06K* OR G01M* OR Y02D*)) OR (CPC =
 (A63F* OR G02B* OR G01N* OR B23Q* OR A61B* OR G06F* OR G06N* OR G06Q* OR G06V* OR
 G10L* OR G16H* OR H04L* OR H04N* OR G06T* OR G06K* OR G01M* OR Y02D* OR G10H* OR
 H04Q*)))
 AND
 (PUD > 2020)

Figure 1: Patent Search Query

A multi-step validation process ensured both recall and precision

- Known GenAI patents were used as benchmarks. Missing entries prompted the inclusion of synonyms and newer model names.
- False positives (e.g., "GAN" as gallium nitride, "transformer" as electrical equipment) were filtered using exclusion terms and proximity constraints.

- Classification-only searches (e.g., G06N 3/0475) were cross-checked against keyword results to identify overlooked patents.
- Manual inspection guided refinement, ensuring coverage without excessive noise.

The modular design of the search (by GenAI subtype) allowed clarity and flexibility during iterative improvement.

Despite a structured and validated search design, several limitations remain common in large-scale patent retrieval:

Terminology Drift. New terms (e.g., “foundation models”) may be absent; older patents may lack standard phrasing.

Inconsistent Labeling. Some patents describe GenAI without using familiar terms (e.g., “dual-network system” for GAN).

Classification Issues. Misclassification or generic codes can cause relevant patents to be missed.

Semantic Ambiguity. Generic terms (e.g., “generator,” “transformer”) may retrieve unrelated patents.

Platform Constraints. Espacenet has syntax and query-length limitations that affect search expressiveness.

While no retrieval method is perfect, this approach provides a transparent and reproducible strategy optimized for GenAI. The final dataset includes 1,570 complete patent records retrieved from Espacenet (2021–2025). XML files were parsed to extract 16 relevant fields out of 20, with the title, abstract, and claims used for topic modeling. These fields offer concise descriptions of the core innovation, while the description field was excluded due to verbosity and frequent boilerplate content. Records missing an abstract or at least one claim were excluded, reducing the dataset from 4,001 to 1,570 high-quality documents. The final dataset offers a domain-specific, well-structured foundation for topic modeling and innovation analysis in GenAI.

4. Methodology

The methodology focuses on applying topic modeling to uncover latent thematic structures within patent documents and to track their evolution over time. A comparative modeling approach was employed to evaluate the performance of multiple topic modeling algorithms on the same dataset, ensuring robustness and consistency in the results.

4.1 Data Preprocessing

Due to their unstructured nature, patent documents require careful preprocessing to support meaningful topic modeling. Standard preprocessing steps include lowercasing, punctuation removal, tokenization, lemmatization, filtering of stopwords and low-frequency terms (Friedl et al., 2013). While stemming is often used in traditional text mining pipelines (Janani & Vijayarani, 2016), this study employed lemmatization instead. Lemmatization preserves grammatical structure and semantic meaning by reducing words to their base forms (e.g., "generating" to "generate") while maintaining syntactic integrity, an important consideration in technical texts. In contrast, stemming can produce non-standard or truncated forms (e.g., "generation" to "gener"), potentially reducing model interpretability. Building on this foundation, a domain-adapted, multi-stage preprocessing pipeline was applied to GenAI patent data. The input corpus was formed by merging the title, abstract, and claims fields, which contain the core technical content. The preprocessing steps included:

Normalization: All text was converted to lowercase, known model names (e.g., GPT-4, DALL·E) were standardized using regular expressions, hyphens were replaced with spaces, and standalone numbers, punctuation, and numeric references were removed to reduce formatting artifacts.

Entity Cleaning: A curated list of named entities was used to retain only domain-relevant technical concepts (e.g., "transformer," "autoencoder").

Stopword Removal: A custom stopwords list was developed by combining domain knowledge with data-driven techniques. Frequent non-informative words were first identified using Counter frequency distributions. Common legal and technical terms such as "apparatus," "method," and "processor" appeared frequently but lacked thematic relevance. To further refine the list, TF-IDF analysis was used to identify low-value terms with minimal discriminative power. Words with low TF-IDF scores were reviewed and, those found to be semantically generic or legally repetitive were added to the stopwords list.

Linguistic Filtering: Part-of-speech tagging was applied to eliminate generic verbs (e.g., "using," "providing") and adjectives (e.g., "new," "present") that lacked semantic specificity.

Tokenization and Lemmatization: Cleaned text was tokenized, and lemmatization was performed using SpaCy to reduce words to their base forms while preserving semantic integrity.

Further refinement included identifying multi-word expressions using Gensim's Phrases model. Frequent domain-specific phrases (e.g., "neural network") were merged into single tokens (e.g., "neural_network") based on co-occurrence and frequency thresholds. Extremely common terms appearing in over 30% of documents were also removed to improve topic distinctiveness and reduce semantic overlap, following standard topic modeling practices (Blei et al., 2003; Sievert & Shirley, 2014). This helped the model focus on more informative and distinctive vocabulary. The resulting text was used to build Gensim dictionaries and bag-of-words corpora, providing the structured input required for topic modeling. Given the legal and technical nature of patent texts, often filled with repetitive

phrasing, boilerplate language, and low-information terminology, this multi-step, domain-specific pipeline was essential for ensuring thematic clarity. Stopword removal extended beyond generic English stopwords to include industry-specific and legal terms frequently found in patent documents. Part-of-speech tagging filtered out non-informative verbs, adjectives and auxiliaries, while tokenization structured the cleaned content into lexical units suitable for modeling. Without these preprocessing steps, early LDA runs produced poor-quality topics dominated by generic keywords, offering little semantic value. In contrast, the finalized pipeline significantly improved the distinctiveness, coherence and interpretability of the resulting topics by retaining only meaningful and thematically rich vocabulary.

4.2 Topic Modelling Approach

Four topic modeling algorithms—LDA, LSI, NMF, and BERTopic—were applied to the dataset for exploratory comparison, following the comparative modeling approach outlined by Arosemena, Tzavellos, and Nanda (2025). To optimize model performance and ensure fair comparison, a grid search was conducted for each algorithm to explore multiple hyperparameter combinations. This systematic tuning helped identify the best-performing configurations based on coherence and diversity scores.

LDA/LSI

Latent Dirichlet Allocation (LDA) was employed as the primary probabilistic topic modeling technique due to its capacity to uncover latent thematic structures from unstructured text while offering high interpretability. As a generative model, LDA assumes each document is a mixture of latent topics, with each topic represented as a distribution over words governed by Dirichlet priors. This formulation supports sparsity, meaning that only a subset of topics dominates each document—an assumption well-suited for domain-specific corpora such as patents (Abdelrazek et al., 2023). To

identify the optimal LDA configuration, a grid search was performed across three key parameters: the number of topics (k), the Dirichlet prior for document-topic distribution (α), and the Dirichlet prior for topic-word distribution (β). The number of topics was varied from 1 to 20 to evaluate different levels of thematic granularity. A six-topic model was ultimately selected, balancing semantic coherence and topic distinctiveness, and meeting the interpretability requirements of innovation-focused patent analysis. For each combination of parameters, coherence scores (C_v) were calculated to assess semantic quality (see Section 5.1). Alpha values ranged from 0.01 to 0.91, covering both symmetric and asymmetric priors. Lower α values encouraged topic sparsity, which is desirable for modeling focused technical content. Similarly, β was varied across the same range, where lower values promote sharper topic-word distributions and help highlight more distinctive keywords (Gupta et al., 2022). Topic diversity was also evaluated to ensure that selected models yielded not only coherent but also non-redundant topic sets. This dual-metric evaluation helped ensure that the final model produced semantically meaningful and interpretable topics suitable for qualitative analysis. To benchmark LDA, Latent Semantic Indexing (LSI) was also implemented. LSI applies singular value decomposition (SVD) to the term-document matrix to project documents and terms into a lower-dimensional latent space, revealing patterns of term co-occurrence. While LSI is effective at capturing synonymy and latent associations, its utility for topic modeling is limited by its reliance on Gaussian assumptions and the absence of a generative probabilistic framework. Moreover, interpretability is often reduced, as the components do not consistently align with human-understandable topics (Abdelrazek et al., 2023). For LSI, additional hyperparameters were tested—specifically, `one_pass` (True/False), `power_iterations` (1–7), and `extra_samples` (50–250)—which influence the accuracy and stability of the truncated SVD. Topic coherence was calculated for each configuration to enable fair comparison with LDA. Topic modeling and related text transformations were implemented using widely recognized Python libraries

for natural language processing and unsupervised learning. The consistency in experimental design across both models supports an objective assessment of their relative performance.

NMF

An extensive parameter tuning procedure was conducted to evaluate the performance of Non-negative Matrix Factorization (NMF). NMF is an algebraic topic modeling technique that decomposes the document-term matrix into two non-negative matrices, producing additive components that improve interpretability. In contrast to the probabilistic nature of LDA, NMF extracts distinct parts-based representations (Abdelrazek et al., 2022). Multiple configurations were tested by varying the number of optimization passes (5 and 10), the kappa parameter (ranging from 0.05 to 1.0), and the maximum number of iterations for the W and H matrices (50, 150, 250). These parameters affect the convergence behavior and stability of the factorization process. Coherence scores were computed for each configuration to identify the most semantically consistent topic structures. Results were stored and ranked to determine the most coherent and interpretable NMF setup.

BERTopic

BERTopic builds on recent advances in transformer-based language models to overcome the semantic limitations of conventional topic modeling techniques like LDA and NMF, which rely on bag-of-words representations and disregard contextual word meanings. By leveraging document-level embeddings (e.g., BERT), BERTopic captures semantic relationships and clusters similar documents using HDBSCAN, a density-based algorithm (Grootendorst, 2022). In this study, BERTopic was applied using two input granularities to explore how abstraction level affects topic representation: (1) BERTopic-chunk, which segmented the patent text into smaller chunks (e.g., paragraph-level) to capture fine-grained thematic variation, and (2)

BERTopic-summary, which used full, merged title-abstract-claim texts per patent to extract broader semantic patterns. This design choice was motivated by the limited context window of transformer-based language models like MiniLM, which typically truncate input beyond 256 tokens. To address this, BERTopic-chunk segmented each patent document into shorter, paragraph-level chunks, ensuring that important semantic details were preserved and not lost during embedding. In contrast, BERTopic-summary used the full, merged title-abstract-claim content to capture broader, document-level semantics. This dual configuration allowed for comparison between fine-grained and holistic topic representations, and reflects established strategies in the literature for handling long input texts in transformer-based topic modeling. For both settings, document vectors were generated using MiniLM embeddings and clustered using KMeans. A parameter search was conducted over the number of clusters (2–20) and Maximal Marginal Relevance (MMR) diversity (0.2–1.0) to identify optimal configurations. Topic representations were extracted via a class-based TF-IDF (c-TF-IDF) approach (Qaiser & Ali, 2018) using a CountVectorizer with 1–2 n-gram ranges. Importantly, unlike other models in this study, BERTopic did not apply frequency-based filtering or remove generic terms during preprocessing, allowing for richer contextual interpretation. Final models were evaluated using topic coherence and diversity to assess their interpretability and distinctiveness. This embedding-based clustering approach overcomes the limitations of bag-of-words models by capturing contextual word meanings, enabling BERTopic to produce semantically rich and thematically distinct topic representations. Unlike models such as LDA, NMF, or LSI, which work with term frequencies and typically remove very common words to reduce noise, BERTopic retains all words during preprocessing. This is because it relies on dense vector representations that learn meaning from context, where even frequent terms can carry important semantic information. Removing these words could reduce the model’s ability to fully capture the relationships between concepts. Additionally, BERTopic does not use Gensim’s corpus structure. Instead, it

applies the CountVectorizer function from scikit-learn to build its vocabulary and extract n-gram features, supporting richer and more flexible text representation.

5. Results and Analysis

5.1 Quantative Evaluation

A comparative evaluation was conducted to address the first research question concerning which topic modeling algorithms generate the most coherent and practically interpretable topics when applied to Generative AI patent data. Quantitative performance was assessed using coherence and topic diversity metrics, while qualitative interpretability was examined through keyword analysis and topic intrusion validation.

To ensure comparability, all models were applied to the same preprocessed dataset under consistent settings. Performance was evaluated using three key quantitative indicators: topic coherence, topic diversity, and overall topic quality. These metrics were selected based on established evaluation practices reviewed by Abdelrazek et al. (2023), as each captures a distinct dimension of interpretability and model utility. **Topic coherence** is a primary indicator of interpretability, assessing whether the top-ranked words in a topic cohere semantically. As defined by Newman et al. (2010), coherence is often measured using Pointwise Mutual Information (PMI), which quantifies the likelihood of word co-occurrence beyond chance. This metric approximates human judgment, with higher scores indicating that a topic contains terms that frequently appear together in the corpus and thus represent a meaningful semantic group. In accordance with prior studies and best practices, coherence was computed based on the top 10 tokens per topic—balancing interpretability, consistency, and computational efficiency. Abdelrazek et al. (2023). While topic coherence measures the internal consistency of individual topics, **topic diversity** assesses how distinct the topics are from one another. It is calculated as the proportion of unique words across the top 10 words of each topic relative to the total

number of words considered. This metric captures the extent to which the model generates non-redundant, thematically differentiated outputs. Although Abdelrazek et al. (2023) originally define topic diversity using the top 25 words per topic, the current study adopts a top-10 word scope to ensure methodological consistency with the coherence evaluation. A high diversity score indicates that the model successfully identifies a broad range of concepts with minimal vocabulary overlap, an essential quality in patent analysis, where topic redundancy can reduce the analytical value of the results. Conversely, a low diversity score suggests that topics may be reusing similar terms, signaling insufficient conceptual separation or overfitting to dominant lexical patterns (Abdelrazek et al., 2023). To capture both interpretability and distinctiveness in a single measure, a composite **topic quality** score was calculated. This score was based on the same top-10 word threshold used in the coherence and diversity evaluations to ensure consistency across metrics. While the 10-topic configuration did not produce the highest individual scores for coherence or diversity, it offered a practical balance between the two, avoiding extremes. As noted by Abdelrazek et al. (2023), coherence and diversity often exist in tension: increasing the number of topics typically enhances diversity but can lower coherence, whereas fewer topics may improve coherence while yielding overly broad or overlapping themes. In this study, the 10-topic solution was selected not for maximizing a single metric, but for aligning with the broader objective of topic quality, generating topics that are both interpretable and sufficiently distinct, even if not optimal on either metric alone.

The final evaluation results are presented in Table 1, which compares the top-performing configurations of five topic modeling algorithms based on their coherence, diversity, and composite quality scores. The table shows that the BERTopic models, particularly the chunk-based variant, demonstrated the highest overall topic quality, while traditional methods, such as LDA and NMF achieved moderate performance. These results reinforce the importance of considering both coherence and diversity when

Models	Topic numbers	Topic Coherence	Topic Diversity	Topic Quality
LDA	5	0.45	0.95	0.43
LSI	5	0.32	0.4	0.13
NMF	7	0.35	0.74	0.26
BERTopic chunks	5	0.74	0.86	0.65
BERTopic summeries	5	0.76	0.83	0.63

Table 1: Final evaluation of topic models with highest topic quality scores

selecting models, as no single metric alone provides a complete picture of topic interpretability. In addition to the final evaluation table, the internal performance dynamics of the five models are visualized in Figures 2-6. Each figure shows how topic coherence, diversity and overall quality change with the number of topics. For LDA (Figure 2), coherence slightly improves up to 10 topics but then remain relatively stable, while topic quality peaks at 5 and declines with more topics, indicating diminishing distinctiveness. This suggests that 6–10 topics strike the best balance between interpretability and uniqueness. In contrast, LSI shows stable coherence but a steady drop in diversity and overall quality as topic count increases, implying reduced distinctiveness despite consistent interpretability.

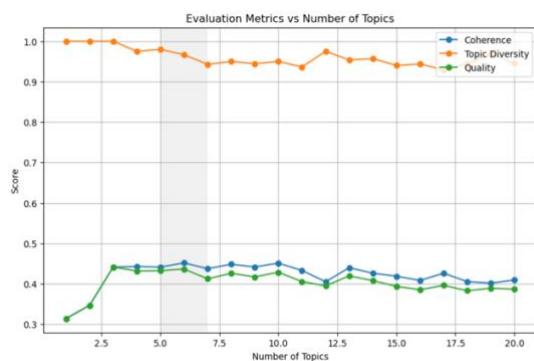


Figure 2: LDA Model Performance

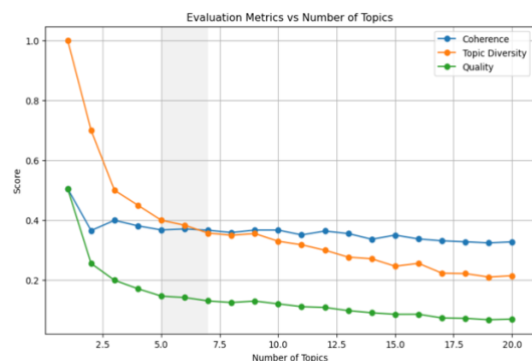


Figure 3: LSI Model Performance

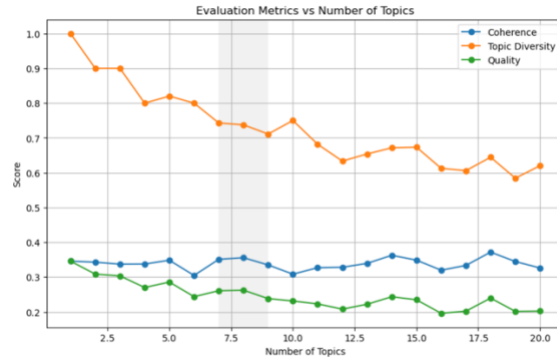


Figure 4. NMF Model Performance

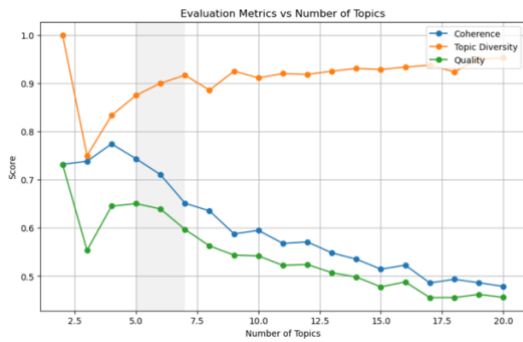


Figure 5: BERTopic chunks

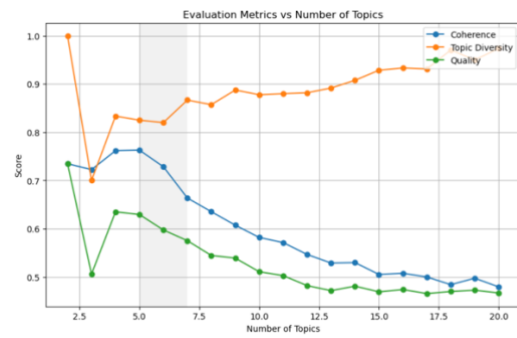


Figure 6: BERTopic summaries

These results suggest that LSI performs best with a small number of broad themes, with topic quality peaking at 5 topics. For NMF, coherence remains stable, but diversity decreases as topic count increases. Topic quality peaks at 7 topics, indicating that NMF performs best when extracting a moderate number of distinct, semantically rich themes. BERTopic (chunk-level) shows high initial coherence that declines with more topics, while diversity remains consistently strong (above 0.9), suggesting well-separated topics even at higher counts. This indicates that while BERTopic-chunk maintains high topic diversity across all settings, the decline in coherence with more topics reduces overall quality, showing that diversity alone is not sufficient. Nonetheless, BERTopic-chunk performs strongest at lower topic counts, offering a well-balanced output of semantically rich and distinct themes. For BERTopic-summary, although diversity remains high, coherence drops more sharply with additional topics, leading to a steeper decline in topic quality.

5.2 Qualitative Evaluation

Three models emerged as top performers in terms of topic quality: BERTopic (chunk), BERTopic (summary), and Latent Dirichlet Allocation (LDA). Although BERTopic summary ranked third in overall topic quality, its qualitative outputs raised interpretability concerns. Despite high quantitative scores, several BERTopic (chunk) topics showed semantic overlap and limited clarity. Terms like *image*, *medical*, and *ultrasound* appeared across multiple topics (e.g., Topics 0, 3, 5), while generic terms such as *model*, *parameter*, and *user* reduced thematic distinctiveness. Some keywords (e.g., *datum*, *loss*) were vague or non-informative. This highlights the importance of complementing metric-based evaluations with qualitative assessment to ensure interpretability. Similarly, the BERTopic chunk model generated granular topic groupings, but its interpretability was limited when evaluated qualitatively due to recurring semantic redundancy and abstract vocabulary. As illustrated in Appendix B, Table 4, for instance, Topic 1 (*text*, *sentence*, *document*, *language*) and Topic 3 (*prompt*, *service*, *query*, *language*) shared overlapping language related to natural language input, yet lacked clear thematic differentiation. Additionally, broad and non-specific terms such as *node*, *item*, *content*, and *performance* appeared across several topics, undermining conceptual clarity. Redundant verbs like *generate*, *learn*, and *training* were also common across clusters, further blurring thematic boundaries. Despite maintaining high topic diversity, the BERTopic chunk model produced topics that were conceptually weak and difficult to interpret, making it unsuitable for subsequent thematic analysis in its current form. In contrast, LDA demonstrated greater semantic coherence and clearer topic separation, as shown in Appendix C, Table 5. Each topic was manually labeled through qualitative analysis of the top-ranked keywords. For example, Topic 0 (*feature*, *sample*, *layer*, *generator*, *discriminator*, *generative adversarial*) was interpreted as **Adversarial Image Generation**, reflecting adversarial training methods such as GANs, with application-oriented terms like *face*,

recognition, and *detection* suggesting relevance to visual recognition systems. Topic 1, **Language Modeling and Prompt-Based Learning**, contained terms pointing to transformer-based NLP systems and instruction tuning. Topic 2, **Sequence Encoding and Audio Signal Processing**, included *signal*, *sequence*, *audio*, *decoder*, *block*, and *quantization*, indicating a focus on sequential data and multimedia encoding architectures. Topic 3 was labeled **Medical Imaging and Diagnostic AI**, based on the presence of terms like *medical*, *sensor*, *segmentation*, *pixel*, and *magnetic resonance*. Topic 4 focused on **Embedded Systems and Signal Transmission**, involving low-level AI deployment contexts, while Topic 5 reflected **Representation Learning and Autoencoding Methods**, with terms such as *vector*, *autoencoder*, *encoder*, and *latent* indicating unsupervised learning and compression techniques. Although BERTopic models achieved slightly higher composite topic quality scores, the decision was made to adopt LDA for the main interpretation and analysis phase. This choice was driven not only by metric considerations but also by LDA’s practical advantages in transparency and interpretability. Unlike BERTopic, which lacks native support for probabilistic topic distributions across documents¹, LDA provides per-document topic probability vectors. This structure allows for further validation through techniques such as topic intrusion tests and semantic plausibility checks, which help verify that topic assignments are meaningfully grounded in the text.

Moreover, the topics generated by LDA demonstrated stronger thematic consistency, with less term redundancy and greater alignment with recognizable technological areas. Given the objective of this thesis to trace the thematic directions of Generative AI innovation rather than to optimize performance metrics alone, LDA offers a more methodologically robust and interpretable foundation. Therefore, it was selected as the final model for conducting topic-based analysis in the remainder of the study.

¹ https://maartengr.github.io/BERTopic/getting_started/distribution/distribution.html#example

LDA Topic Validation through Topic Intrusion Analysis

To assess whether the LDA-generated topics aligned with human interpretability, a topic intrusion task was conducted, following the method introduced by Chang et al. (2009). For each patent, the model-generated topic distribution was used to identify the top three most probable topics, alongside a randomly selected low-probability *intruder* topic. A short piece from the patent abstract was presented along with the four topic labels, and the task was to identify the topic that did not semantically fit. If the model's topic assignments are meaningful, the intruder should be clearly identifiable (Chang et al., 2009). In most cases, the intruder topic was easily recognized, suggesting strong alignment between the model-generated topics and the underlying patent content. One illustrative example further highlights this interpretability: patent EP22791474A1², which describes a vibration energy harvesting monitoring device, was assigned three relevant topics, all of which contained terms highly consistent with the patent's technical focus. In contrast, the intruder topic contained generic language processing terms such as *text*, *sentence*, *query*, and *language*, which has no semantic relevance to the patent. This clear thematic mismatch made the intruder easily distinguishable (see Figure 7). Similar outcomes were observed across the sample, consistently supporting the interpretability and coherence of the LDA model's output. To complement the manual evaluation, a Topic Log Odds (TLO) analysis was also performed to systematically evaluate topic coherence at scale. TLO is defined as the log ratio of the probability mass assigned by the model to the true intruder topic versus the probability mass assigned to the topic selected as the intruder by a human subject (Chang et al., 2009).

$$\text{TLO} = \log_{10}\left(\frac{P_{\text{model-intruder}}}{P_{\text{human-selected-intruder}}}\right)$$

² <https://patentscope.wipo.int/search/en/WO2022224698>

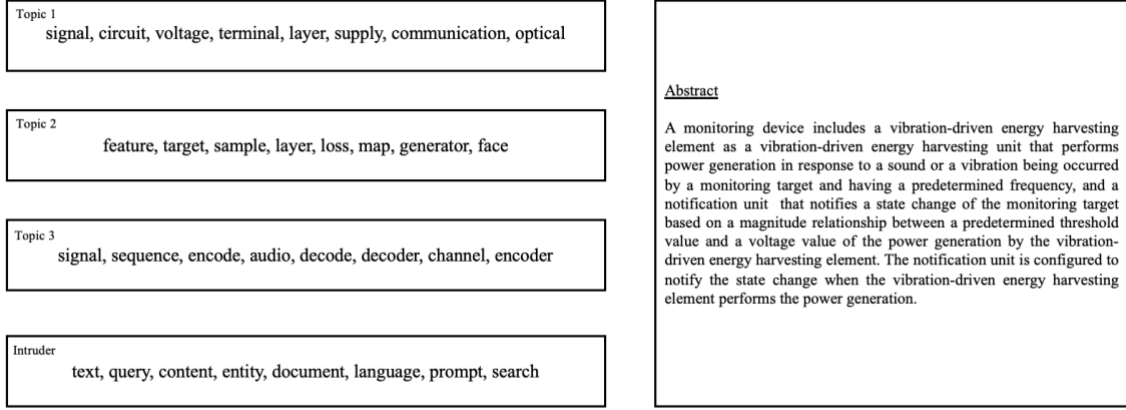


Figure 7: Sample Topic Intrusion Test

This formulation captures how well human judgments align with the model's assessment of irrelevant topics. In the present study, a fully automated version of the TLO metric was implemented to enable large-scale evaluation across hundreds of patent documents without requiring human intervention. This version modifies the original formulation in two key ways: first, instead of asking humans to select the intruder, a predefined intruder topic was assigned to each document. This intruder was drawn from a different topic cluster, ensuring thematic irrelevance. Second, the third-highest topic assigned by the model was used as a proxy for relevance, under the assumption that it is still semantically related to the document but less dominant than the top two topics.

$$\text{TLO} = \log_{10}\left(\frac{P_{\text{intruder}}}{P_{\text{third-most-relevant-topic}}}\right)$$

More negative values indicate stronger topic coherence, as the model assigns much lower probability to the intruder than to any relevant topic. This adaptation maintains the conceptual intent of the original method, quantifying topic separability, while relying solely on model outputs. Figure 8 shows the distribution of TLO scores across the patent corpus. As expected, more negative values indicate higher topic coherence, reflecting

a greater gap between the probabilities of relevant and intruding topics. The histogram reveals that while many patents cluster near zero, indicating marginal topic separation, a substantial portion of documents fall well below -1 . This tail suggests clear model confidence in rejecting irrelevant topics in those cases. To summarize, 36.56% of all patents show TLO values below -1 , providing evidence that the LDA model produced coherent and well-separated topic structures for a significant share of the dataset.

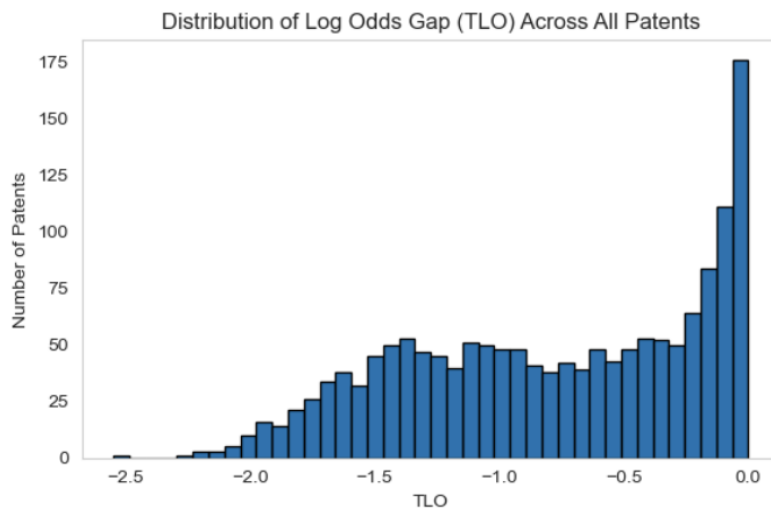


Figure 8: Topic Log Odds

5.3 Thematic Evolution of Generative AI Patents

To address the second research question, a thematic analysis was conducted using the most interpretable topic model identified in the previous section. Latent Dirichlet Allocation (LDA) was selected based on its superior semantic coherence and clearer topic separation compared to alternative models such as BERTopic. Using this model, six distinct innovation themes were extracted. Each patent was then assigned to its most representative topic, allowing for a structured exploration of how Generative AI innovations are thematically distributed and how these themes evolve over time. The number of documents associated with each topic is presented in Figure 9. Topic 5, previously labeled *Representation Learning and Autoencoding Methods*, appeared in 419 patents, making it

the most frequent innovation theme. It was followed by Topic 1, *Language Modeling and Prompt-Based Learning*, with 291 patents, indicating a strong focus in these two areas. Topics 0, 3, and 4 also had moderate representation, covering themes such as *Adversarial Image Generation*, *Medical Imaging and Diagnostic AI*, and *Embedded Systems*. In contrast, Topic 2, related to *Audio Signal and Sequence Encoding*, had the lowest number of documents ($n = 174$), suggesting a more niche area of innovation.

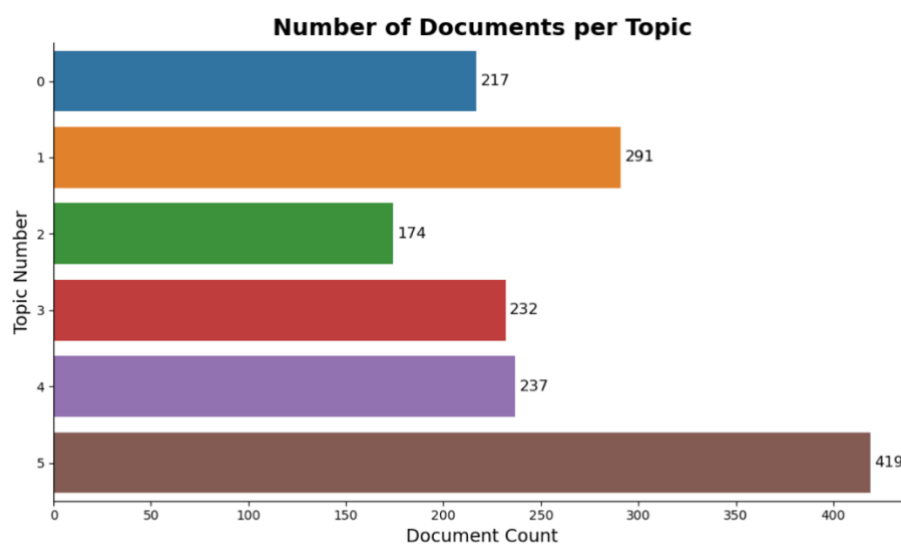


Figure 9: Number of Documents per Topic

One limitation of Figure 9 is that it assigns each document to a single dominant topic, which oversimplifies the multi-thematic nature of patent texts. In contrast, the full topic distribution table reveals that most documents contain a blend of multiple themes in varying proportions. For example, Document 1 consists of 27% Topic 0, 47% Topic 2, and 18% Topic 4, rather than belonging exclusively to one. This demonstrates that topic modeling captures the thematic heterogeneity within patents, something that network analysis, which typically relies on binary citation or co-authorship links, cannot fully reflect. While network-based methods help identify structural relationships, they often treat nodes and links statically and fail to capture the content-based richness of technological

development. Topic modeling, on the other hand, provides a probabilistic and continuous representation of how multiple innovation themes co-occur and evolve within a single document, offering a more nuanced and fine-grained understanding of innovation dynamics. The topic distribution table (see Figure 10) illustrates how individual patents often span multiple innovation themes. For instance, Document 1 combines elements of Topic 2, Topic 0, and Topic 4, suggesting convergence between generative model development, signal processing architectures, and system-level optimization. This thematic layering highlights the complexity of recent Generative AI innovations.

	Topic_0	Topic_1	Topic_2	Topic_3	Topic_4	Topic_5
0	0.62%	26.68%	10.32%	0.94%	1.42%	60.03%
1	27.47%	1.07%	47.48%	0.60%	18.15%	5.23%
2	0.85%	49.92%	20.08%	17.29%	0.93%	10.93%
3	0.25%	92.31%	0.25%	0.31%	0.36%	6.53%
4	0.52%	0.56%	0.49%	78.58%	12.72%	7.13%
5	17.06%	2.23%	9.72%	3.62%	2.48%	64.88%
6	0.24%	97.83%	0.30%	0.45%	0.51%	0.67%
7	11.43%	0.48%	47.14%	24.19%	0.41%	16.34%
8	11.50%	30.46%	0.19%	32.00%	12.57%	13.27%
9	28.54%	0.38%	12.86%	15.64%	7.34%	35.25%

Figure 10: Topic distributions for the first 10 documents

Compared to earlier patents, which often show a single dominant topic, later documents reflect increased interdisciplinarity, potentially signaling a shift from foundational methods toward integrated applications. Such evolution is difficult to capture using static network methods, reinforcing the value of topic modeling in uncovering fine-grained innovation trajectories. To observe how innovation themes evolve, a yearly topic share analysis was conducted. Each year, the share of patents dominated by each topic was computed. As shown in Figure 11, Topic 1 demonstrates a strong upward trend, particularly from 2023 to 2025, indicating increasing

research and commercial attention. In contrast, Topic 5, while initially dominant, appears to be stabilizing or slightly declining in recent years.

Other topics, such as Topics 4 and 3, show moderate fluctuation but no strong growth. The share of patents held by the top 5 applicants was analyzed to understand which organizations are driving innovation. Figure 12 shows how the share of patents held by the top five applicants in Generative AI changes over time, from 2021 onward. This helps us understand how different companies have increased, decreased, or maintained their focus on GenAI innovation during this period. Koninklijke Philips N.V. shows a clear increase in patent activity after 2022, which shows a growing investment in GenAI technologies. Analyzing topic assignments by applicant reveals that Koninklijke Philips N.V. is most active in Topic 3, accounting for over 60% of its patents in the dataset. Based on the top keywords and thematic labeling of Topic 3 (see Section 5.2), this theme corresponds to “Medical Imaging and Diagnostic AI”. This finding aligns closely with Philips’ known domain expertise and strategic focus in healthcare technologies³. It also demonstrates how topic modeling can successfully uncover the thematic priorities of different applicants, linking technological themes to institutional actors.

The company’s secondary contributions to Topic 5 (“Representation Learning and Autoencoding”) and Topic 1 (“Language Modeling and Prompt-based Learning”) suggest additional areas of exploration but with significantly lower intensity. Overall, the results show that Philips is not just an active filer in GenAI, but a leading innovator in healthcare-related AI applications, reinforcing the model’s interpretability and its value in mapping thematic trajectories across firms. On the other hand, Samsung Electronics Co., Ltd., which had a strong presence early on, shows a drop in patent share after 2021. This could mean a temporary slowdown or a shift in focus to other areas. At first glance, this reduction might suggest a slowdown in innovation activity. However, a closer look at the topic

³ [Philips Healthcare](#)

modeling results reveals a more nuanced explanation. Despite the drop in overall share, Samsung remains an active contributor in specific thematic areas. In particular, the company's patents are heavily concentrated in Topic 2 and Topic 4, which together account for the majority of its filings. Based on the top keywords associated with these topics, Topic 2 appears to focus on signal processing, sequence modeling, and audio generation, while Topic 4 relates to embedded AI systems, hardware integration, and sensor-based applications. These areas are consistent with Samsung's strategic interests in consumer electronics, smart devices, and multimedia AI technologies, suggesting that the decline in overall patent share does not indicate withdrawal but rather a refinement of scope. Instead of broad diversification, Samsung may be investing more deeply in product-specific innovations that integrate GenAI functionalities into core business areas. This pattern highlights the value of combining quantitative trend analysis with thematic topic modeling. While raw patent counts may indicate shifts in activity, topic-level insights offer a more accurate view of where innovation is actually happening. In Samsung's case, the data suggests not a decline in innovation intensity but rather a strategic pivot toward specialized GenAI applications aligned with its existing product ecosystem. Robert Bosch GmbH shows a steady increase in patent activity through 2023, indicating a consistent and growing commitment to GenAI innovation. Tata Consultancy Services Limited maintains a strong and stable presence, reflecting sustained engagement. Siemens AG contributes at a steady but lower level, suggesting a more cautious or narrowly focused strategy. These trends highlight how some companies actively expand their innovation efforts while others take a more reserved approach.

A lifecycle analysis was carried out for each of the six LDA topics further to interpret the thematic structure of Generative AI innovation. This analysis aimed to determine the maturity stage of each theme—such as whether it is emerging, growing, or stabilizing. Two key metrics were used: the average yearly patent share (to assess a topic's overall presence) and the Compound Annual Growth Rate (CAGR), which captures the average

annual rate of change over time. Following the method proposed by Ghaffari et al. (2024), the CAGR was calculated based on the topic's patent share in the first and final year of the observation period. Together, these measures help distinguish between well-established and rapidly expanding innovation areas, offering a more dynamic view of how thematic priorities evolve over time.

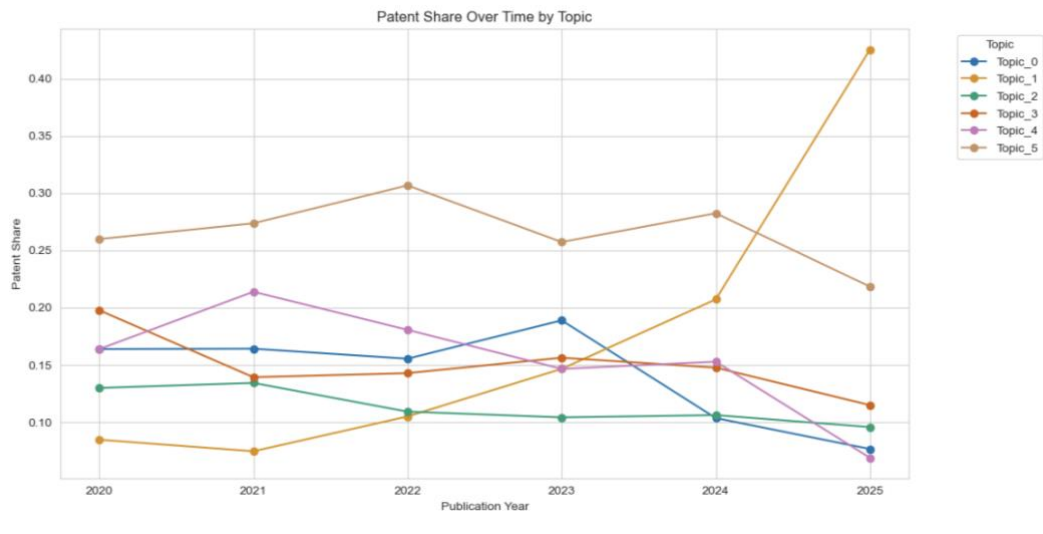


Figure 11: Patent Share over time

This approach identifies topic lifecycles in terms of patent share growth, enabling the classification of innovation themes as *emerging*, *dominant*, *saturating*, or *declining* over time (Ghaffari et al., 2024). Figure 13 synthesizes this time series data into a two-dimensional lifecycle classification. The horizontal axis reflects the average topic share, capturing overall dominance, while the vertical axis captures the CAGR, indicating the direction and intensity of change. Four quadrants emerge:

- Top-right (High Share, High Growth): Emerging and expanding themes
- Bottom-right (High Share, Negative Growth): Mature or saturated themes
- Top-left (Low Share, High Growth): Niche but growing areas
- Bottom-left (Low Share, Negative Growth): Declining themes

In this framework, **Topic 1** is positioned in the *emerging* phase, characterized by a relatively low average patent share but a high positive growth rate, indicating increasing recent attention. Although Topic 1 holds the second-highest total document count, its average annual share over the entire period remains moderate due to limited early presence. Combined with a high positive CAGR in recent years as shown in Table 2, this positions Topic 1 in the *emerging* quadrant of the lifecycle matrix, reflecting its growing presence in recent patent filings.

Topic	Label/Theme	Avg Share	CAGR	Zone
Topic_5	Representation Learning and Autoencoding Methods	0.266391	-0.034194	Saturated
Topic_1	Language Modeling and Prompt-Based Learning	0.173923	0.380743	Emerging
Topic_4	Embedded Systems and Signal Transmission	0.154473	-0.158912	Declining
Topic_3	Medical Imaging and Diagnostic AI	0.149811	-0.102825	Declining
Topic_0	Adversarial Image Generation	0.142111	-0.141	Declining
Topic_2	Sequence Encoding and Audio Signal Processing	0.113292	-0.059175	Declining

Table 2: Lifecycle Classification of GenAI Topics

Note: Topics are categorized into lifecycle based on their average annual share of patents and Compound Annual Growth Rate (CAGR) from 2021 to 2025. This classification highlights the relative maturity and dynamism of each innovation theme.

In contrast, Topic 5 lies in the *saturation* phase, maintaining a high overall share but showing low or slightly negative growth, suggesting it may be reaching maturity. Topics 0, 3, and 4 are classified as declining themes. While each had a moderate presence across the dataset, their consistently negative growth rates indicate a downward trend in innovation activity, suggesting they may be reaching saturation or are being deprioritized in the GenAI landscape. For example, the most frequently occurring IPC code in Topic 4 is A61B 18/12, which pertains to surgical instruments that deliver energy—such as high-frequency current—to body tissue. This indicates that earlier GenAI innovation efforts were concentrated in embedded or energy-based medical applications. The observed decline in patent filings under this code supports the broader downward trend identified in Topic 4, potentially reflecting technological saturation or a strategic shift away from this application area toward newer

GenAI landscape. Lastly, Topic 2 appears relatively *stable*, maintaining low levels of growth. The results of the topic modeling and lifecycle analysis provide a structured view of the dominant and emerging innovation trajectories in Generative AI patent filings. Among the six identified themes, Topic 5, which relates to *Representation Learning and Autoencoding Methods*, holds the highest average share of patents, suggesting it has been the most actively explored domain in recent years. The most frequently associated IPC classification within this topic is G06N 3/04, covering convolutional neural networks (CNNs), a foundational architecture in deep learning. This indicates that a substantial portion of GenAI patent activity has focused on core neural model development and representation learning. However, despite its dominance, the theme exhibits a slightly negative growth rate, implying that innovation in this space may be reaching saturation, with diminishing returns or fewer breakthroughs in recent years.

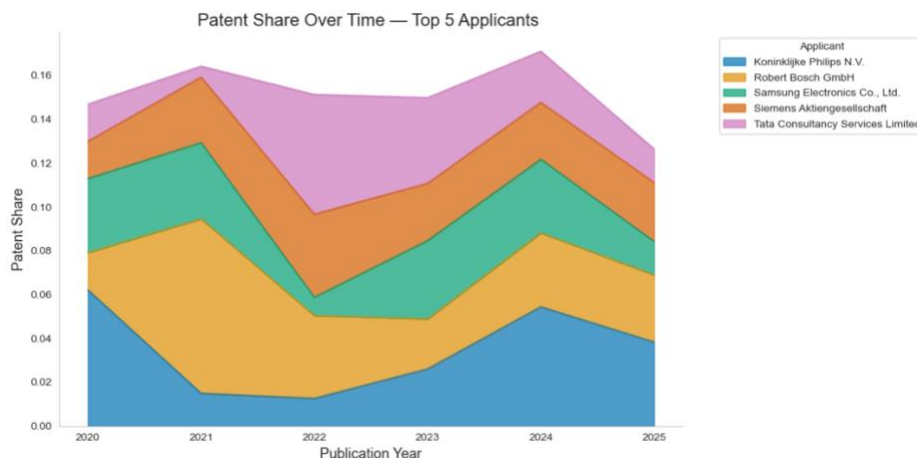


Figure 12: Patent Share Over Time — Top 5 Applicants

In contrast, Topic 1, labeled as *Language Modeling and Prompt-Based Learning*, demonstrates both strong growth and an increasing annual share of patent activity. While it does not hold the highest average share across the entire period, its recent acceleration places it firmly in the emerging quadrant of the topic lifecycle matrix, signaling a rapidly expanding area of strategic importance. The most frequently occurring IPC code in this theme

is G06N 20/00, which covers machine learning systems that adapt based on models and data—directly aligning with the rise of large language models (LLMs) and prompt-based architectures. This convergence suggests that Topic 1 is not only gaining traction but may also become a central pillar of GenAI innovation, particularly as instruction tuning, fine-tuning, and deployment of foundational models continue to evolve.

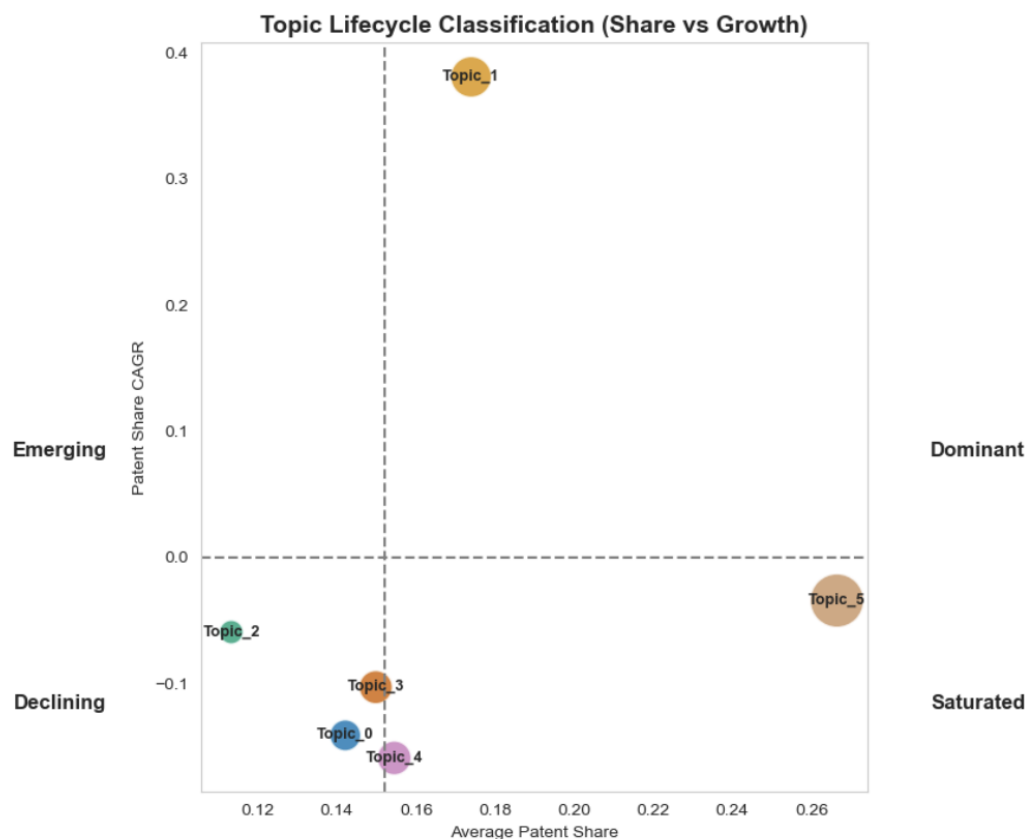


Figure 13: Topic Lifecycle Classification (Average Share vs. Growth)

In summary, the topic lifecycle analysis reveals distinct maturity levels across innovation themes in the GenAI patent landscape. While some areas like *Language Modeling and Prompt-Based Learning* are gaining strategic relevance, others such as *Embedded Systems* and *Medical Imaging* show signs of decline or stabilization. By combining average patent share and growth dynamics, this framework offers a nuanced view of technological evolution, highlighting where attention is intensifying, plateauing, or fading. These insights provide a valuable foundation for

understanding not only where innovation is happening, but also how priorities are shifting over time across different domains of Generative AI.

5.4 Mapping Innovation Stages in Generative AI

A DoI-Theoretical Framework

To address Research Question 3—*how thematic insights inform strategic innovation decisions within the Generative AI ecosystem*—this section applies the DoI framework to assess the maturity and strategic positioning of identified topic. The DoI framework (Rogers, 2003) segments innovation adoption into five categories: Innovators, Early Adopters, Early Majority, Late Majority, and Laggards. By linking these stages to topic-level patent trends, the study identifies which themes are emerging, maturing, or declining. Each topic was evaluated using indicators such as Compound Annual Growth Rate (CAGR), average patent share, unique applicant count, and IPC code diversity. These serve as proxies for innovation lifecycle stages and support a systematic classification of topics. Figure 14 shows applicant and technological (IPC) diversity per topic. Topics 4 and 5 attract the broadest applicant base, suggesting widespread industrial interest, while Topics 0 and 2 are more narrowly adopted. Topic 4 also exhibits the widest IPC spread, indicating high cross-domain potential. In contrast, Topics 2 and 3 show narrower diffusion, potentially reflecting niche or underexplored areas. Figure 15 illustrates temporal adoption patterns. Topic 1 (Language Modeling and Prompt-Based Learning) displays rapid growth post-2022, aligning with the early adopter phase. Conversely, Topic 5 (Representation Learning and Autoencoding Methods) shows stable high activity, suggesting maturity. Topics 3 and 4, exhibit early peaks followed by decline, suggest movement into the late majority. Topic 2 remains consistently low, signaling limited diffusion.

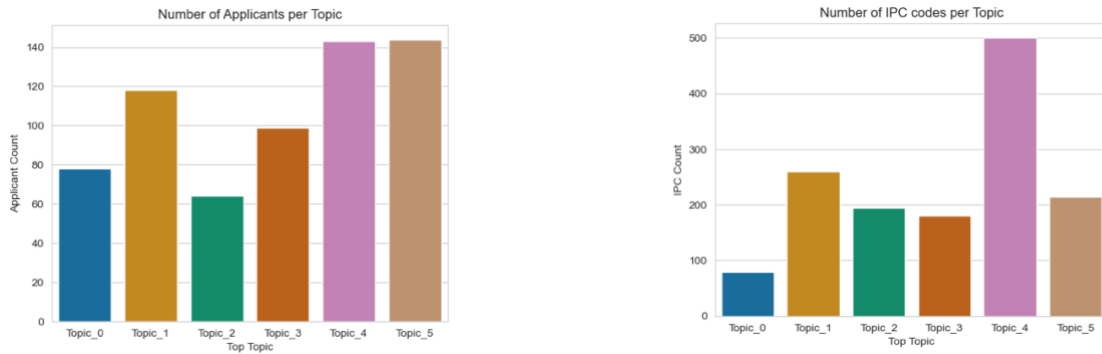


Figure 14: Applicant and Classification Diversity

By linking Figures 14 and 15 to the Diffusion of Innovations (DoI) framework, the analysis offers actionable insights into the maturity and spread of each GenAI innovation theme, addressing RQ3. Figure 14 highlights applicant and technological diversity across topics, Topics 4 and 5 show broad sectoral adoption, while Topic 2 appears niche and underexplored. Figure 15 maps temporal adoption trends, positioning themes along the innovation lifecycle. Topic 1 (Language Modeling and Prompt-Based Learning) shows rapid growth, suggesting early-stage momentum and strategic potential, whereas Topic 5 exhibits a plateau, indicating maturity and potential saturation. To deepen this analysis, applicant-level diffusion was explored within Topic 1. The cumulative patent shares of Philips N.V. were mapped to DoI stages using IPC-applicant pairs, classifying their progression as Innovators, Early Adopters, Early Majority, Late Majority, or Laggards. This helped contextualize how key players adopt and shape emerging technologies over time. This approach follows the standard Diffusion of Innovations (DoI) framework (Rogers, 2003), which segments adopters into five groups along a bell-shaped curve. Cumulative adoption thresholds were applied as follows: $\leq 5\%$ as Innovators, 5–16% as Early Adopters, 16–50% as Early Majority, 50–84% as Late Majority, and $>84\%$ as Laggards. These thresholds are grounded in Rogers' theory and reflect typical adoption patterns.

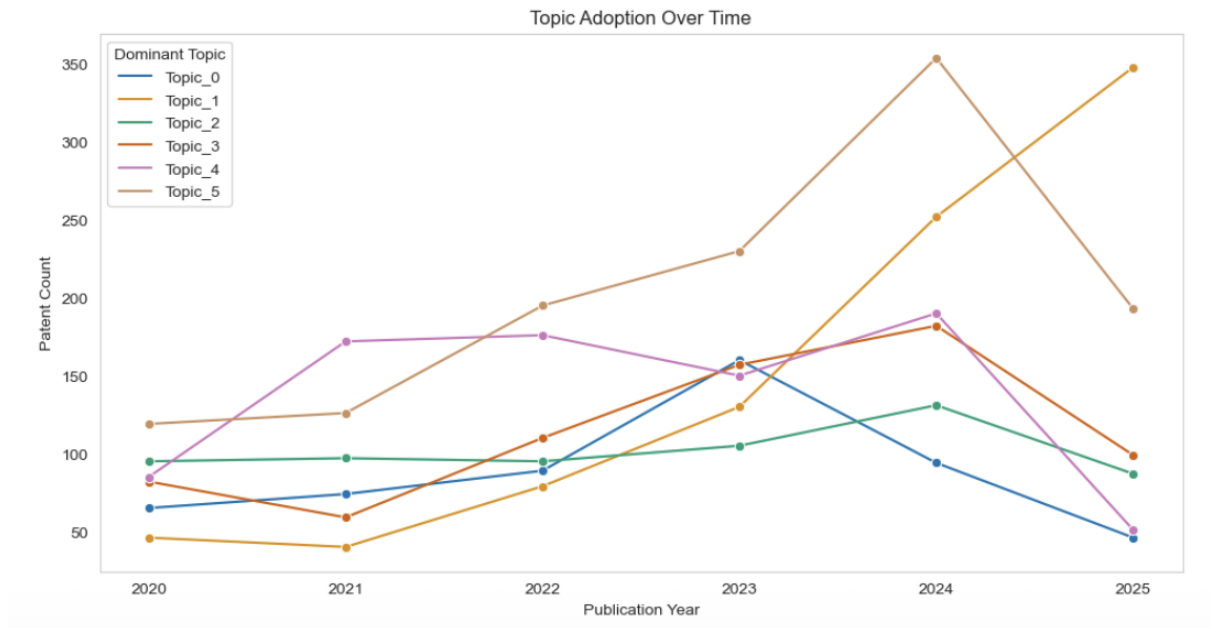


Figure 15: Patent Adoption Trends by Topic

Descriptive statistics further validate the segmentation. For instance, Innovators have cumulative shares around 2.5–5%, Early Adopters average 10.7%, Early Majority 33.2%, Late Majority 67.3%, and Laggards 92.6%. This consistency supports the framework’s applicability to GenAI patent diffusion.

As shown in Figure 16, Philips N.V. demonstrates adoption across stages, from early engagement (e.g., G05B 15/00 in 2025) to broader scaling (e.g., G06N 3/02 in 2024). This structured progression illustrates the company's strategic shift from experimentation toward mainstream adoption in GenAI. While this section focuses on lifecycle dynamics for Philips within Topic 1, the same approach can be applied to other topics and applicants. To maintain clarity, only one illustrative case is presented. In 2025, Philips held a 4% cumulative share in IPC code G05B 15/00, classifying it as an Innovator—suggesting early experimentation with GenAI technologies. Three additional codes (G06F 3/01, G06F 40/10, G06F 40/30) also appeared in 2025 within the Early Adopter range (5–16%), reflecting structured exploration of transformer-based and prompt-driven applications. By 2024, Philips reached the Early Majority stage for G06N 3/02, indicating broader integration of GenAI into its portfolio.

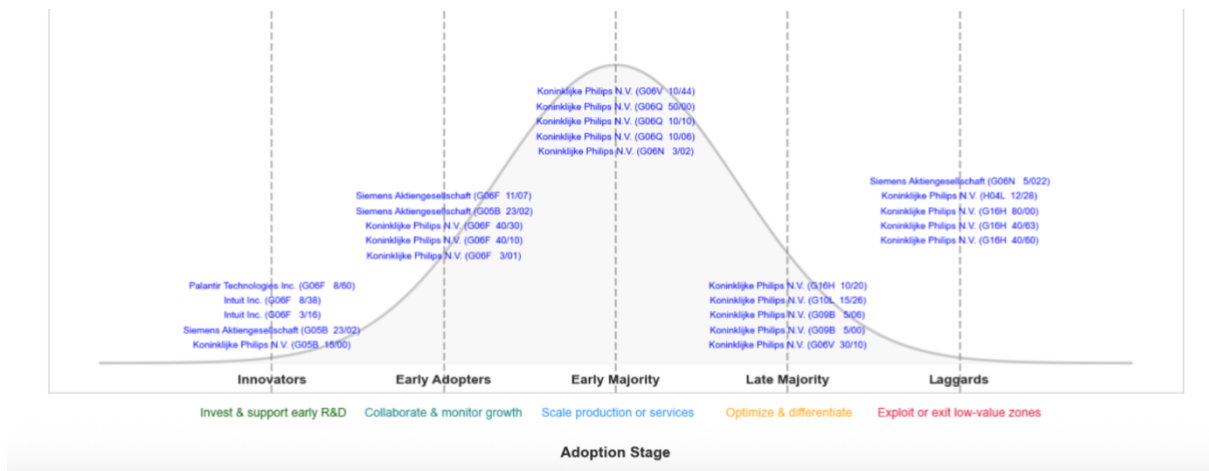


Figure 16. Adoption stage

This micro-level analysis demonstrates that within a single organization, adoption can span multiple lifecycle stages depending on technological focus. It also highlights the strategic evolution from experimentation to scale. Interpreted through the DoI prespective, these patterns offer insight into both the timing and nature of GenAI adoption, supporting RQ3 by showing how topic-level patent data can inform innovation strategy, identify early-stage collaboration opportunities, and uncover underexplored areas for investment.

6. Discussions and Conclusions

This thesis set out to investigate the thematic and temporal dynamics of innovation in Generative AI using a large-scale patent dataset and a combination of topic modeling and diffusion analysis. Six distinct technological themes were identified, each reflecting different stages of maturity, adoption and strategic relevance. By applying multiple topic modeling algorithms and validating results both quantitatively and qualitatively, the analysis revealed a shift from multimodal and embedded systems toward more text-centric, prompt-driven innovations. Using the

Diffusion of Innovations (DoI) framework, applicants were mapped across lifecycle stages, identifying early adopters. These insights offer practical guidance for strategic decision-making on investment and collaboration priorities. To contextualize the findings, this thesis was qualitatively compared with Singh et al. (2024). Despite methodological differences, both studies found similar themes, including medical AI, Image Generation and processing, conversational agents. This study extended Singh et al.'s analysis by using a broader set of models (LDA, LSI, NMF, BERTopic) and providing more granular adoption insights via patent trend analysis and CAGR-based lifecycle classification. The present study also employed multi-topic distribution analysis, offering a finer-grained perspective compared to the single-label approach in STM and network-based studies. This method captured hybrid and cross-domain innovations more effectively, a valuable feature in a rapidly evolving domain like Generative AI. While the findings demonstrate strong thematic alignment with existing literature, this thesis contributes additional value through algorithmic benchmarking, more detailed diffusion insights, and an updated dataset covering recent developments post-2023. Despite these contributions, several limitations should be acknowledged. First, patent data may not fully capture informal or open-source innovation pathways that are particularly relevant in the AI domain. Second, the use of IPC codes and applicant names as proxies for technological and organizational behavior introduces abstraction that may overlook qualitative nuances. Lastly, the diffusion analysis assumes a relatively linear adoption model, which may not fully capture the feedback loops and discontinuities characteristic of AI innovation. Future work could extend this study in multiple directions. Incorporating citation networks or inventor collaboration graphs could yield deeper insight into knowledge flows and innovation clusters. Additionally, integrating external datasets, such as research publications, venture capital activity or open-source contributions, would help validate the findings and address the limitations of patent-only analysis. Moreover, although BERTopic was preprocessed differently, specifically without removing part-of-speech tags to preserve

contextual richness, its qualitative outputs still suggest that further refinement is needed. Adopting a more context-preserving strategy could enhance topic coherence and interpretability, better aligning with the model's transformer-based architecture. Overall, this thesis provides an evidence-based foundation for understanding the evolving structure of Generative AI innovation and contributes a novel methodological approach for translating patent data into strategic foresight.

References

Arosemena, G., Tzavellos, F., & Nanda, R. (2025). *Mapping the meaning of human dignity at the European Court of Human Rights: An unsupervised learning approach* [Manuscript in preparation]. Law and Tech Lab, Faculty of Law, Maastricht University.

Abdelrazek, A., Eid, Y., Gawish, E., Medhat, W., & Hassan, A. (2023). Topic modeling algorithms and applications: A survey. *Information Systems*, 112, 102131. <https://doi.org/10.1016/j.is.2022.102131>

Blei, D. M., & Lafferty, J. D. (2009). *Topic models*. In A. Srivastava & M. Sahami (Eds.), *Text Mining: Classification, Clustering, and Applications* (pp. 71–94). Chapman and Hall/CRC. <https://aclanthology.org/W14-3110/>

Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent Dirichlet allocation. *Journal of Machine Learning Research*, 3, 993–1022. <https://dl.acm.org/doi/10.5555/944919.944937>

Chang, J., Boyd-Graber, J., Gerrish, S., Wang, C., & Blei, D. (2009). Reading tea leaves: How humans interpret topic models. *Advances in Neural Information Processing Systems*, 22, 288–296. https://www.researchgate.net/publication/221618226_Reading_Tea_Leaves_How_Humans_Interpret_Topic_Models

Choe, H., Lee, D. H., Seo, I. W., & Kim, H. D. (2013). Patent citation network analysis for the domain of organic photovoltaic cells: Country, institution, and technology field. *Renewable and Sustainable Energy Reviews*, 26, 492–505. <https://doi.org/10.1016/j.rser.2013.05.037>

Dernis, H., Calvino, F., Moussiegt, L., Nawa, D., Samek, L., & Squicciarini, M. (2023). *Identifying artificial intelligence actors using online data*.

OECD. <https://www.researchgate.net/publication/369362298>

Dhingra, S., Singh, M., SB, V., Malviya, N., & Gill, S. S. (2023). *Mind meets machine: Unravelling GPT-4's cognitive psychology*. arXiv.

<https://arxiv.org/abs/2303.11436>

Diesner, J., & Carley, K. (2010, August). A methodology for integrating network theory and topic modeling and its application to innovation diffusion. In *2010 IEEE Second International Conference on Social Computing* (pp. 687–692). IEEE.

<https://doi.org/10.1109/SocialCom.2010.106>

Duch, J., & Arenas, A. (2005). Community detection in complex networks using extremal optimization. *Physical Review E*, 72(2), 027104.

<https://doi.org/10.1103/PhysRevE.72.027104>

Dwivedi, Y. K., Sharma, A., Rana, N. P., Giannakis, M., Goel, P., & Dutot, V. (2023). Evolution of artificial intelligence research in *Technological Forecasting and Social Change*: Research topics, trends, and future directions. *Technological Forecasting and Social Change*, 192, 122579.

<https://doi.org/10.1016/j.techfore.2023.122579>

European Patent Office. (2022). *Espacenet – pocket guide*. European Patent Office. <https://link.epo.org/web/technical/espacenet/espacenet-pocket-guide-en.pdf>

European Patent Office. (2025, February 24). *Why switch to the current Espacenet version?* European Patent Office.

<https://www.epo.org/en/searching-for-patents/helpful-resources/patent-knowledge-news/why-switch-current-espacenet-version>

European Patent Office. (n.d.). *CPC Text Categoriser*.

<https://epn.epo.org/cpc-text-categoriser>

European Patent Office. (n.d.). *Expert services*.

<https://data.epo.org/expert-services/>

Feldman, R., & Sanger, J. (2006). The text mining handbook: Advanced approaches in analyzing unstructured data. *Cambridge University Press*.

https://www.researchgate.net/publication/280113623_The_Text_Mining_Handbook_Advanced_Approaches_in_Analyzing_Unstructured_Data

Ghaffari, M., Aliahmadi, A., Khalkhali, A., Zakery, A., Daim, T. U., & Zamani, M. (2024). *Exploring the technological leaders using tire industry patents: A topic modeling approach*. *Technology in Society*, 78, 102664. <https://www.sciencedirect.com/science/article/pii/S0160791X24002124>

Grootendorst, M. (2022). *BERTopic: Neural topic modeling with a class-based TF-IDF procedure*. arXiv. <https://arxiv.org/abs/2203.05794>

Gupta, P., Ding, B., Guan, C., & Ding, D. (2024). Generative AI: A systematic review using topic modelling techniques. *Data and Information Management*, 8(2), 100066. <https://doi.org/10.1016/j.dim.2024.100066>

Gupta, R. K., Agarwalla, R., Naik, B. H., Evuri, J. R., Thapa, A., & Singh, T. D. (2022). Prediction of research trends using LDA based topic modeling. *Global Transitions Proceedings*, 3(1), 298–304. <https://doi.org/10.1016/j.gltp.2022.03.015>

Hall, B. H., & Harhoff, D. (2012). Recent research on the economics of patents. *Annual Review of Economics*, 4, 541–565.

<https://doi.org/10.1146/annurev-economics-080511-111008>

<https://doi.org/10.1016/j.techsoc.2024.102664>

<https://teddykw2.wordpress.com/wp-content/uploads/2012/07/everett-m-rogers-diffusion-of-innovations.pdf>

Janani, R., & Vijayarani, S. (2016). Text mining research: A survey. *International Journal of Innovative Research in Computer and Communication Engineering*, 4(4), 5845–5852.

https://www.researchgate.net/publication/339552018_Text_Mining_Research_A_Survey

Kherwa, P., & Bansal, P. (2019, July). Topic modeling: A comprehensive review. *EAI Endorsed Transactions on Scalable Information Systems*, 6(22), 159623. <https://doi.org/10.4108/eai.13-7-2018.159623>

Krakowski, S. (2025). Human-AI agency in the age of generative AI. *Information and Organization*, 35(1), 100560. <https://doi.org/10.1016/j.infoandorg.2025.100560>

Liu, N., Shapira, P., Yue, X., & Guan, J. (2021). Mapping technological innovation dynamics in artificial intelligence domains: Evidence from a global patent analysis. *PLOS ONE*, 16(12), e0262050. <https://doi.org/10.1371/journal.pone.0262050>

Murray, F., & Stern, S. (2005). Do formal intellectual property rights hinder the free flow of scientific knowledge? An empirical test of the anti-commons hypothesis. *Journal of Economic Behavior & Organization*, 63(4), 648–687. <https://doi.org/10.1016/j.jebo.2006.05.017>

Newman, D., Lau, J. H., Grieser, K., & Baldwin, T. (2010). Automatic evaluation of topic coherence. In *Proceedings of the 2010 Annual Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* (pp. 100–108). Association for Computational Linguistics.

<https://www.researchgate.net/publication/220817098>

Obreja, D., Rughinis, R., & Rosner, D. (2024). Mapping the conceptual structure of innovation in artificial intelligence research: A bibliometric analysis and systematic literature review. *Journal of Innovation & Knowledge*, 9, 100465. <https://doi.org/10.1016/j.jik.2024.100465>

Qaiser, S., & Ali, R. (2018). Text mining: Use of TF-IDF to examine the relevance of words to documents. *International Journal of Computer Applications*, 181(1). <https://doi.org/10.5120/ijca2018917395>

Rogers, E. M. (1962). Diffusion of innovations. *Free Press*.

Rost, K. (2011). The strength of strong ties in the creation of innovation. *Research Policy*, 40(4), 588–604.

<https://doi.org/10.1016/j.respol.2010.12.001>

Saxena, A., & Iyengar, S. (2020). Centrality measures in complex networks: A survey. *arXiv preprint arXiv:2011.07190*.

<https://arxiv.org/abs/2011.07190>

Sengar, S. S., Hasan, A. B., Kumar, S., & Carroll, F. (2024). *Generative artificial intelligence: A systematic review and applications*. arXiv.

<https://arxiv.org/abs/2405.11029>

Shomee, H. H., Wang, Z., Ravi, S. N., & Medya, S. (2024). A comprehensive survey on AI-based methods for patents. *arXiv preprint arXiv:2404.08668*. <https://arxiv.org/abs/2404.08668>

Sievert, C., & Shirley, K. (2014). LDAvis: A method for visualizing and interpreting topics. In J. Chuang, S. Green, M. Hearst, J. Heer, & P. Koehn (Eds.), *Proceedings of the Workshop on Interactive Language Learning*,

Visualization, and Interfaces (pp. 63–70). Association for Computational Linguistics. <https://doi.org/10.3115/v1/W14-3110>

Singh, S., Singh, S., Kraus, S., Sharma, A., & Dhir, S. (2024). Characterizing generative artificial intelligence applications: Text-mining-enabled technology roadmapping. *Journal of Innovation & Knowledge*, 9(3), 100531. <https://doi.org/10.1016/j.jik.2024.100531>

Toetzke, M., Probst, B., & Feuerriegel, S. (2023). Leveraging large language models to monitor climate technology innovation. *Environmental Research Letters*, 18(9), 091004.
https://www.researchgate.net/publication/373290858_Leveraging_large_language_models_to_monitor_climate_technology_innovation

Vijayarani, S., & Janani, R. (2016). Text mining: Open source tokenization tools—An analysis. *Advanced Computational Intelligence: An International Journal*, 3(1), 37–47.
<https://www.scirp.org/reference/referencespapers?referenceid=2393442>

World Intellectual Property Organization. (2023). *Patent landscape report: Generative artificial intelligence (GenAI) — Appendices*. WIPO.
<https://www.wipo.int/web-publications/patent-landscape-report-generative-artificial-intelligence-genai/en/appendices.html>

World Intellectual Property Organization. (n.d.). *Generative AI*. WIPO.
<https://www.wipo.int/en/web/patent-analytics/generative-ai>

Zhang, H., Daim, T., & Zhang, Y. (2021). Integrating patent analysis into technology roadmapping: A latent Dirichlet allocation-based technology assessment and roadmapping in the field of blockchain. *Technological Forecasting and Social Change*, 167, 120729.
<https://doi.org/10.1016/j.techfore.2021.120729>

Zhou, Y., Liu, J., & Kröger, P. (2024). Unlocking the potential of generative AI for patent analysis (*arXiv preprint arXiv:2406.19465*).
<https://doi.org/10.48550/arXiv.2406.19465>

Appendices

Appendix A

Topic 0	Topic 1	Topic 2	Topic 3	Topic 4
face	synthetic	user	medical image	feature
image	learning	prompt	medical	pose
face image	learning model	query	segmentation	product
target	dataset	language	imaging	fingerprint
image image	machine	question	anatomical	depth
loss	machine learning	llm	image	pattern
adversarial	model	content	ultrasound	image datum
area	variable	language model	tissue	light
adversarial network	training	platform	patch	object
object	instance	answer	chromosome	point cloud

Table 3: Representative terms across the 5 topics from the BERTopic summary model

Appendix B

Topic 0	Topic 1	Topic 2	Topic 3	Topic 4
parameter	text	circuit	prompt	generator
learning	sentence	voltage	query	loss
model	graph	terminal	user	discriminator
dataset	entity	switch	language	adversarial
datum	document	connect	llm	image
design	language	supply	language model	training image
machine	similarity	transformer	service	adversarial network
learning model	label	capacitor	question	generative adversarial
synthetic	tag	couple	server	real
machine learning	user	electrode	large language	training
learn	character	winding	large	network
training	embed	signal	platform	generative
set	textual	electric	natural language	object
series	word	inductor	natural	sample image
training datum	content	connected	client	map
variable	vector	battery	content	depth image
generative	node	primary	management	depth

Table 4: Representative terms across the 5 topics from BERTopic chunk model

Appendix C

Topic 0	Topic 1	Topic 2	Topic 3	Topic 4	Topic 5
feature	text	signal	medical	signal	vector
target	query	sequence	sensor	circuit	representation
sample	content	encode	dimensional	voltage	parameter
layer	entity	audio	segmentation	terminal	dataset
loss	document	decode	space	layer	machine_learn
map	language	decoder	region	supply	label
generator	prompt	cannel	pixel	communication	score
face	search	encoder	machine	optical	latent
vector	sentece	block	patch	detection	class
discriminator	message	video	color	drive	synthetic
generative_adversarial	llm	frame	machine_learn	pattern	classification
point	generative	parameter	object	transmit	autoencoder
inputte	Natural_language	coefficient	location	sensor	node
vehicle	agent	prediction	interest	light	encoder
region	database	frequency	position	radio_frequeny	generative
parameter	file	residual	magnetic_resonance	controller	processor
learn	candidate	\noise	environment	sense	algorithm
weight	task	sound	patient	line	product
recognition	visual	quantization	display	side	model
detection	token	speaker	anatomical	energy	learn

Table 5: Representative terms across the 6 topics from LDA model

Official statement of original thesis

By signing this statement, I hereby acknowledge the submitted thesis (hereafter mentioned as "product"), titled:

Decoding Innovation Dynamics in Generative AI: A Patent-Based Topic Modelling and Diffusion Theory Approach

.....
to be produced independently by me, without external help.

Wherever I paraphrase or cite literally, a reference to the original source (journal, book, report, internet, etc.) is given.

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First and last name: Lilit Poghosyan

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Study programme: Business Intelligence & Smart Services

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Course/skill: Master Thesis EBS4040-EBS4047

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Statement on the use of Generative AI (GenAI) in the master thesis

I hereby certify that I adhered to the SBE guidelines on the use of GenAI tools such as ChatGPT in the master thesis. In the box below, I document how and for what purposes I used GenAI.

During the preparation of this work, I used GenAI for the following purposes:

- Search engine: [Google, EPO; used to locate peer-reviewed articles and patents related to Generative AI innovations.]
- Ideation helper: [ChatGPT; Assisted in brainstorming research questions, structuring analysis sections and exploring theoretical frameworks like Diffusion of Innovations.]
- Text summarizer: [ChatGPT; used for summarizing long articles and patent abstracts into concise descriptions during topic interpretation.]
- Explanation provider: [ChatGPT; provided technical explanations of machine learning algorithms (e.g., LDA, BERTopic) and helped interpret modeling outputs.]
- Language assistant: [Grammarly, ChatGPT; supported grammar refinement, clarity improvements and academic tone adjustments during writing and editing phases.]
- Table editor: [Microsoft Word; tables were formatted using Word]
- Translator: [Google Translate; used to translate foreign-language patent titles or abstracts when full English versions were unavailable.]
- Other: [Visualization generator; Python libraries; used for creating figures.]

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Course/skill: Master Thisis EBS4040-EBS4047

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ID number:

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Signature:

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Sustainable Development Goals (SDG) Statement

Name Lilit Poghosyan
ID
Supervisor Rohan Nanda
Date 02.07.2025

Through the research conducted for this master's thesis, I seek to contribute to one or more of the 17 SDG(s) set forth by the United Nations (<https://www.undp.org/sustainable-development-goals>). Specifically:



SDG Code(s): 9

Explanation (max. 300 words): This study supports SDG 9 by advancing knowledge on innovation trends in Generative AI. Using topic modeling and patent analysis, it identifies key technological themes and their evolution over time. These insights help policymakers and industry leaders invest strategically in high-potential areas, supporting the development of resilient and future-ready innovation infrastructure.

The research also promotes more inclusive industrial growth by highlighting opportunities beyond dominant regions and sectors, encouraging support for emerging technologies and smaller firms. By tracking the lifecycle of AI innovations, the study enables early identification of shifts, allowing infrastructure and policy to adapt proactively. In doing so, it bridges data-driven analysis with strategic planning, contributing to a more sustainable and equitable innovation ecosystem.